3.32 Formal Power Series and Laurent Series

Definition 3.32.1 (Formal Laurent Series)
*Let A be a commutative ring. Define the **formal Laurent series ring** $A((X))$ to be the set of sequences $(c_n)_{n\in\mathbb{Z}}$ in A such that there exists N for which $n \le N \implies c_n = 0$. Addition is defined pointwise with product defined as follows*

$$\left(\sum_{n\in\mathbb{Z}} c_n X^n\right)\left(\sum_{n\in\mathbb{Z}} d_n X^n\right) = \sum_{n\in\mathbb{Z}}\left(\sum_{k\in\mathbb{Z}} c_{n-k}d_k\right)X^n$$

where the inner sum is a sum of a family with all but finitely many terms zero. That this is well-defined and makes $A((X))$ into a commutative ring is shown in (3.32.3).

Definition 3.32.2 (Order of a Formal Laurent Series)
*Let A be a commutative ring. For $0 \neq h = \sum_n c_n X^n \in A((X))$ define the **order** $v(h) := \min\{n \mid c_n \neq 0\}$, and $v(0) := \infty$. The minimum exists since the c_n vanish for n sufficiently negative and $h \neq 0$. We call $c_{v(h)}$ the **leading coefficient** of h.*

Proposition 3.32.3 ($A((X))$ is a Commutative Ring)
Let A be a commutative ring and $c, d \in A((X))$. For every $n \in \mathbb{Z}$ the family $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ has all but finitely many terms zero, so $(cd)_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ is defined, and $(cd)_n = 0$ for $n < v(c) + v(d)$. In particular $cd \in A((X))$. Moreover $A((X))$ is a commutative ring under pointwise addition and this product.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The product is well-defined. Let $c = (c_n)_{n\in\mathbb{Z}}$ and $d = (d_n)_{n\in\mathbb{Z}}$ be elements of $A((X))$. If $c = 0$ or $d = 0$ every term below is zero and the product is 0, so assume $v(c), v(d)$ are finite. Fix $n \in \mathbb{Z}$. If the term $c_{n-k}d_k$ is nonzero then $n - k \ge v(c)$ and $k \ge v(d)$, that is

$$v(d) \le k \le n - v(c)$$

so only finitely many $k \in \mathbb{Z}$ give a nonzero term. Hence $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ has all but finitely many terms zero, and the inner sum $e_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ is defined by (3.2.50). Moreover if $n < v(c) + v(d)$ then there is no integer k with $v(d) \le k \le n - v(c)$, so every term vanishes and $e_n = 0$. Therefore $(e_n)_{n\in\mathbb{Z}}$ is a sequence in A vanishing for $n < v(c) + v(d)$, that is an element of $A((X))$.

Ring axioms. Write $(cd)_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ for the coefficients of the product, an element of $A((X))$ by the first part.

Addition. For $c, d \in A((X))$ we have $c_n + d_n = 0$ and $-c_n = 0$ for $n < \min(v(c), v(d))$, so $c + d$ and $-c$ lie in $A((X))$, as does the zero sequence. So $A((X))$ is a subgroup of the abelian group $A^{\mathbb{Z}}$ under pointwise addition.

Identity. Let $1 := (\delta_{n,0})_{n\in\mathbb{Z}}$, which vanishes for $n \le -1$ and so lies in $A((X))$. For $c \in A((X))$ the family $(\delta_{n-k,0}c_k)_{k\in\mathbb{Z}}$ has $k = n$ as its only possibly nonzero term, so $(1c)_n = c_n$.

Commutativity. The map $k \mapsto n - k$ is a bijection of $\mathbb{Z}$, so by (3.2.53) and commutativity of A,

$$(cd)_n = \sum_{k\in\mathbb{Z}} c_{n-k}d_k = \sum_{k\in\mathbb{Z}} c_k d_{n-k} = \sum_{k\in\mathbb{Z}} d_{n-k}c_k = (dc)_n$$

Distributivity. For $c, d, d' \in A((X))$, the families $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ and $(c_{n-k}d'_k)_{k\in\mathbb{Z}}$ have all but finitely many terms zero, so by (3.2.51)

$$(c(d+d'))_n = \sum_{k\in\mathbb{Z}} c_{n-k}(d_k + d'_k) = \sum_{k\in\mathbb{Z}} c_{n-k}d_k + \sum_{k\in\mathbb{Z}} c_{n-k}d'_k = (cd)_n + (cd')_n$$

and the other distributive law follows by commutativity.

Associativity. Let $c, d, e \in A((X))$ and fix $n \in \mathbb{Z}$; if any of them is 0 both sides are 0, so assume $v(c), v(d), v(e)$ are finite. Set $g(j,k) := c_{n-j-k}d_j e_k$ for $(j,k) \in \mathbb{Z}\times\mathbb{Z}$. If $g(j,k) \neq 0$ then $j \ge v(d)$, $k \ge v(e)$ and $n - j - k \ge v(c)$, so $v(d) \le j \le n - v(c) - v(e)$ and $v(e) \le k \le n - v(c) - v(d)$; hence g has all but finitely many terms zero. Recall that $(cd)_m = \sum_{j\in\mathbb{Z}} c_{m-j}d_j$ and $(de)_m = \sum_{k\in\mathbb{Z}} d_{m-k}e_k$. Then

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$$\begin{aligned}((cd)e)_n &= \sum_{k\in\mathbb{Z}} (cd)_{n-k}e_k && \text{by definition of the product}\\ &= \sum_{k\in\mathbb{Z}}\left(\sum_{j\in\mathbb{Z}} c_{n-k-j}d_j\right)e_k && \text{by definition of } (cd)_{n-k}\\ &= \sum_{k\in\mathbb{Z}}\sum_{j\in\mathbb{Z}} g(j,k) && \text{by (3.3.17)}\\ &= \sum_{(j,k)\in\mathbb{Z}\times\mathbb{Z}} g(j,k) && \text{by (3.2.53)}\end{aligned}$$

and on the other side

$$\begin{aligned}(c(de))_n &= \sum_{l\in\mathbb{Z}} c_{n-l}(de)_l && \text{by definition of the product}\\ &= \sum_{l\in\mathbb{Z}} c_{n-l}\sum_{k\in\mathbb{Z}} d_{l-k}e_k && \text{by definition of } (de)_l\\ &= \sum_{l\in\mathbb{Z}}\sum_{k\in\mathbb{Z}} g(l-k,k) && \text{by (3.3.17)}\\ &= \sum_{(l,k)\in\mathbb{Z}\times\mathbb{Z}} g(l-k,k) && \text{by (3.2.53)}\\ &= \sum_{(j,k)\in\mathbb{Z}\times\mathbb{Z}} g(j,k) && \text{by (3.2.53) for the bijection } (l,k)\mapsto(l-k,k)\end{aligned}$$

Hence $((cd)e)_n = (c(de))_n$ for all n. ∎

Lemma 3.32.4 (Valuation of a Sum and a Product)
Let A be a commutative ring and $c, d \in A((X))$. Then $v(1) = 0$, $v(-c) = v(c)$, and

a) $v(c+d) \geq \min(v(c), v(d))$
b) $v(cd) \geq v(c) + v(d)$
c) if A is an integral domain then so is $A((X))$, and $v(cd) = v(c) + v(d)$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Clearly $v(1) = 0$ and $v(-c) = v(c)$. If $n < \min(v(c), v(d))$ then $c_n = d_n = 0$, so $(c+d)_n = 0$, which gives a). For b), if $c = 0$ or $d = 0$ then $cd = 0$ and both sides are ∞. Otherwise $(cd)_n = 0$ for $n < v(c) + v(d)$ by (3.32.3), which means $v(cd) \geq v(c) + v(d)$. For c), let $c, d \neq 0$ with $m := v(c)$ and $r := v(d)$. A term $c_{m+r-k}d_k$ of $(cd)_{m+r}$ is nonzero only if $k \geq r$ and $m + r - k \geq m$, that is $k = r$, so $(cd)_{m+r} = c_m d_r \neq 0$ as A is a domain, while $(cd)_n = 0$ for $n < m + r$. Hence $cd \neq 0$ and $v(cd) = m + r$. ∎

Definition 3.32.5 (Formal Power Series)
*Let A be a commutative ring. Define the **formal power series ring** $A[[X]]$ to be the subring of $A((X))$ consisting of those sequences $(c_n)_{n\in\mathbb{Z}}$ for which $n < 0 \implies c_n = 0$, or equivalently those $h \in A((X))$ for which $v(h) \geq 0$.*

Proof that $A[[X]]$ is a subring of $A((X))$. Drafted by Claude Sonnet, reviewed by David Rufino.

$A[[X]] = \{h \in A((X)) \mid v(h) \geq 0\}$, so by (3.32.4) it contains 1 and is closed under negation, addition and multiplication. ∎

Lemma 3.32.6 (Formal Power Series Rings are Separated)
Let A be a commutative ring. For $N \geq 0$ we have $X^N A[[X]] = \{h \in A[[X]] \mid v(h) \geq N\}$, and $\bigcap_{N\geq 0} X^N A[[X]] = 0$.

Proof. If $h = X^N h'$ with $h' \in A[[X]]$ then the coefficients of h below X^N vanish, so $v(h) \geq N$. Conversely if $v(h) \geq N$ then $h = X^N h'$ where h' is h with the coefficients shifted down by N, an element of $A[[X]]$. If h lies in every $X^N A[[X]]$ then $v(h) \geq N$ for every N, so every coefficient of h vanishes and $h = 0$. ∎

Definition 3.32.7 (Formal Derivative)
Let A be a commutative ring. For $f = \sum_n c_n X^n \in A((X))$, define $f' := \sum_n n c_n X^{n-1} \in A((X))$, that is $(f')_n = (n+1)c_{n+1}$.

Lemma 3.32.8 (Valuation of a Derivative)
Let A be a commutative ring and $f \in A((X))$. Then $v(f') \geq v(f) - 1$. In particular if $f \in A[[X]]$ then $f' \in A[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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If $n < v(f) - 1$ then $n + 1 < v(f)$, so $c_{n+1} = 0$ and $(f')_n = (n+1)c_{n+1} = 0$. This gives $v(f') \geq v(f) - 1$. If $f \in A[[X]]$ then $v(f) \geq 0$ and $(f')_n = 0$ for $n < -1$; also $(f')_{-1} = 0 \cdot c_0 = 0$, so $f' \in A[[X]]$. ∎

Definition 3.32.9 (Summable Family in $A((X))$)
*Let A be a commutative ring. A family $(h_i)_{i\in I}$ in $A((X))$ is **summable** if for every $N \in \mathbb{Z}$ all but finitely many $i \in I$ satisfy $v(h_i) \geq N$. Its **sum** is the formal Laurent series*

$$\sum_{i\in I} h_i := \sum_{n\in\mathbb{Z}} \left(\sum_{i\in I} (h_i)_n \right) X^n$$

where $(h_i)_n$ denotes the coefficient of X^n in h_i.

Proof that the sum is well-defined. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $h_{i,n} := (h_i)_n$ and fix $n \in \mathbb{Z}$. By hypothesis (with $N = n+1$) the set $\{i \in I \mid v(h_i) \leq n\}$ is finite, and it contains every i with $h_{i,n} \neq 0$. So $(h_{i,n})_{i\in I}$ has all but finitely many terms zero and $s_n := \sum_{i\in I} h_{i,n}$ is defined by (3.2.50). Let $J_0 := \{i \in I \mid v(h_i) \leq 0\}$, which is finite, and $N_0 := \min\big(0, \min_{i\in J_0} v(h_i)\big)$ (with $N_0 := 0$ if J_0 is empty). If $n < N_0$ then $v(h_i) > n$ for every i, so every $h_{i,n} = 0$ and $s_n = 0$. Hence $(s_n)_{n\in\mathbb{Z}}$ lies in $A((X))$. ∎

Proposition 3.32.10 (Sums of Summable Families)
Let A be a commutative ring and $(h_i)_{i\in I}$ a summable family in $A((X))$. Then

a) If $v(h_i) \geq N$ for every $i \in I$ then $v\big(\sum_{i\in I} h_i\big) \geq N$. In particular if every $h_i \in A[[X]]$ then $\sum_{i\in I} h_i \in A[[X]]$.
b) For a bijection $\sigma : I' \to I$ the family $\big(h_{\sigma(i')}\big)_{i'\in I'}$ is summable and $\sum_{i'\in I'} h_{\sigma(i')} = \sum_{i\in I} h_i$.
c) The family $(h_i')_{i\in I}$ is summable and $\big(\sum_{i\in I} h_i\big)' = \sum_{i\in I} h_i'$.
d) For any $g \in A((X))$ the family $(gh_i)_{i\in I}$ is summable and $g \cdot \sum_{i\in I} h_i = \sum_{i\in I} gh_i$.
e) If $(k_i)_{i\in I}$ is another summable family then $(h_i + k_i)_{i\in I}$ is summable and $\sum_{i\in I}(h_i + k_i) = \sum_{i\in I} h_i + \sum_{i\in I} k_i$.
f) If $I = \bigsqcup_{\lambda\in\Lambda} I_\lambda$ is a partition of I into disjoint subsets then each $(h_i)_{i\in I_\lambda}$ and $\Big(\sum_{i\in I_\lambda} h_i\Big)_{\lambda\in\Lambda}$ is summable, and $\sum_{i\in I} h_i = \sum_{\lambda\in\Lambda} \sum_{i\in I_\lambda} h_i$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $h_{i,n} := (h_i)_n$ and $s_n := \sum_{i\in I} h_{i,n} = \big(\sum_{i\in I} h_i\big)_n$.

a) If $n < N$ then $h_{i,n} = 0$ for every i, so $s_n = 0$, as $J = \emptyset$ is a valid choice in (3.2.50). Hence $v\big(\sum_{i\in I} h_i\big) \geq N$, and taking $N = 0$ gives the statement for $A[[X]]$.

b) The reindexed family clearly meets the summability condition. The family $(h_i)_{i\in I}$ itself is not almost zero in $A((X))$, but for each n the coefficient family $(h_{i,n})_{i\in I}$ in A is, so the statement holds coefficientwise by (3.2.53).

c) Since $v(h_i') \geq v(h_i) - 1$ by (3.32.8) the family $(h_i')_{i\in I}$ is also summable, so $\sum_{i\in I} h_i'$ is defined. Using $(h')_n = (n+1)h_{n+1}$, for every $n \in \mathbb{Z}$

$$\begin{aligned}
\left(\left(\sum_{i\in I} h_i \right)' \right)_n &= (n+1)s_{n+1} = (n+1) \sum_{i\in I} h_{i,n+1} && \text{by definition of the sum} \\
&= \sum_{i\in I} (n+1) h_{i,n+1} && \text{by (3.3.17)} \\
&= \sum_{i\in I} (h_i')_n = \left(\sum_{i\in I} h_i' \right)_n && \text{by definition of the sum}
\end{aligned}$$

d) Let $g \in A((X))$. If $g = 0$ both sides are 0, so assume $v(g)$ is finite. By (3.32.4) $v(gh_i) \geq v(g) + v(h_i)$, so $(gh_i)_{i\in I}$ is summable and $\sum_{i\in I} gh_i$ is defined. Fix $n \in \mathbb{Z}$ and set $a(k,i) := g_{n-k} h_{i,k}$ for $(k,i) \in \mathbb{Z} \times I$. If $a(k,i) \neq 0$ then $k \leq n - v(g)$ and $v(h_i) \leq k$, so i lies in the finite set $\{i \in I \mid v(h_i) \leq n - v(g)\}$ and $v(h_i) \leq k \leq n - v(g)$; hence a has all but finitely many terms zero. Then

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$$\begin{aligned}
\left(g\sum_{i\in I}h_i\right)_n &= \sum_{k\in\mathbb{Z}} g_{n-k}s_k = \sum_{k\in\mathbb{Z}} g_{n-k}\sum_{i\in I}h_{i,k} && \text{by definition of the product and the sum}\\
&= \sum_{k\in\mathbb{Z}}\sum_{i\in I} a(k,i) && \text{by (3.3.17)}\\
&= \sum_{i\in I}\sum_{k\in\mathbb{Z}} a(k,i) && \text{by (3.2.53)}\\
&= \sum_{i\in I}\sum_{k\in\mathbb{Z}} g_{n-k}h_{i,k} && \text{by definition of } a\\
&= \sum_{i\in I}(gh_i)_n = \left(\sum_{i\in I} gh_i\right)_n && \text{by definition of the product and the sum}
\end{aligned}$$

e) By (3.32.4) a), $v(h_i+k_i)\geq\min(v(h_i),v(k_i))$, so for every N all but finitely many i satisfy $v(h_i+k_i)\geq N$ and $(h_i+k_i)_{i\in I}$ is summable. The identity holds in each coefficient by (3.2.51).

f) The subfamilies are summable. For $N\in\mathbb{Z}$ the set $F_N := \{i\in I \mid v(h_i)<N\}$ is finite, so only finitely many λ have $I_\lambda\cap F_N\neq\emptyset$, and for every other λ we have $v\left(\sum_{i\in I_\lambda}h_i\right)\geq N$ by a). Hence $\left(\sum_{i\in I_\lambda}h_i\right)_{\lambda\in\Lambda}$ is summable. The identity holds in each coefficient by (3.2.52) applied to $(h_{i,n})_{i\in I}$. ∎

Corollary 3.32.11 (Interchange of Summation)
Let A be a commutative ring and $(a(i,j))_{(i,j)\in I\times J}$ a summable family in $A((X))$. Then each family $(a(i,j))_{j\in J}$ and $(a(i,j))_{i\in I}$ is summable, the families $\left(\sum_{j\in J}a(i,j)\right)_{i\in I}$ and $\left(\sum_{i\in I}a(i,j)\right)_{j\in J}$ are summable, and

$$\sum_{(i,j)\in I\times J} a(i,j) = \sum_{i\in I}\sum_{j\in J} a(i,j) = \sum_{j\in J}\sum_{i\in I} a(i,j).$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Apply (3.32.10) f) to the partition of $I\times J$ into the rows $\{i\}\times J$, and then to the partition into the columns $I\times\{j\}$. ∎

Lemma 3.32.12 (Convergence Criterion for a Sum)
Let A be a commutative ring, $(h_j)_{j\geq 0}$ a summable family in $A((X))$ and $h\in A((X))$. Then $h=\sum_{j\geq 0}h_j$ if and only if the partial sums converge to h, that is $v\left(h-\sum_{j\leq J}h_j\right)\to\infty$ as $J\to\infty$ (for every $N\in\mathbb{Z}$ we have $v\left(h-\sum_{j\leq J}h_j\right)\geq N$ for all sufficiently large J).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $S:=\sum_{j\geq 0}h_j$ and fix $N\in\mathbb{Z}$. As the family is summable, there is J_0 with $v(h_j)\geq N$ for all $j>J_0$. For every J, partitioning $\mathbb{N}$ into $\{0,\dots,J\}$ and $\{j>J\}$ gives $S-\sum_{j\leq J}h_j=\sum_{j>J}h_j$ by (3.32.10), and for $J\geq J_0$ this has $v\geq N$ by part a) of that proposition.

If $h=S$ then $h-\sum_{j\leq J}h_j=\sum_{j>J}h_j$ has $v\geq N$ for every $J\geq J_0$, so the partial sums converge to h. Conversely suppose the partial sums converge to h, and choose $J\geq J_0$ with $v\left(h-\sum_{j\leq J}h_j\right)\geq N$. Then by (3.32.4)

$$v(h-S)\geq\min\left(v\left(h-\sum_{j\leq J}h_j\right),v\left(\sum_{j>J}h_j\right)\right)\geq N.$$

This holds for every N, so every coefficient of $h-S$ vanishes. ∎

Proposition 3.32.13 (Units of a Formal Power Series Ring)
Let A be a commutative ring and $u\in A[[X]]$, with constant coefficient u_0 (the coefficient of X^0). Then u is a unit of $A[[X]]$ if and only if $u_0\in A^\star$. In particular, when $A=k$ is a field, $u\in k[[X]]^\star\iff u_0\neq 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose $u_0\in A^\star$ and set $w:=1-u_0^{-1}u\in A[[X]]$. Its constant coefficient is 0, that is $v(w)\geq 1$, so $v(w^i)\geq i$ for all $i\geq 0$ by (3.32.4). Hence $(w^i)_{i\geq 0}$ is summable (3.32.9) and $s:=\sum_{i\geq 0}w^i\in A[[X]]$ by (3.32.10). Then

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$$\begin{aligned} s &= 1 + \sum_{i\geq 1} w^i && \text{splitting off } i=0 \text{ by (3.32.10) } f) \\ &= 1 + \sum_{i\geq 0} w^{i+1} && \text{reindexing } i \mapsto i+1 \text{ by (3.32.10) } b) \\ &= 1 + ws && \text{distributivity, by (3.32.10) } d) \end{aligned}$$

so $(1-w)s = 1$. Since $u_0^{-1}u = 1-w$, this says $u\cdot(u_0^{-1}s) = 1$, so $u_0^{-1}s$ is an inverse of u.

Conversely, if u is a unit with inverse $t \in A[[X]]$ then $(ut)_0 = \sum_{k\in\mathbb{Z}} u_{-k}t_k = u_0t_0$, as $u_{-k} = 0$ for $k > 0$ and $t_k = 0$ for $k < 0$. So $u_0t_0 = 1$ and $u_0 \in A^\star$. When $A = k$ is a field, $u_0 \in k^\star$ is equivalent to $u_0 \neq 0$. ∎

Proposition 3.32.14 (Units of a Formal Laurent Series Ring)
Let A be an integral domain and $0 \neq h \in A((X))$ with $m := v(h)$. The following are equivalent

a) h is a unit of $A((X))$
b) $h = X^m u$ for a unit $u \in A[[X]]^\star$
c) the leading coefficient h_m of h is a unit of A

In particular when $A = k$ is a field, $k((X))$ is a field.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

c) $\implies$ b): $u := X^{-m}h \in A[[X]]$ has constant coefficient $u_0 = h_m \in A^\star$, so u is a unit of $A[[X]]$ by (3.32.13).

b) $\implies$ a): $X^{-m}u^{-1}$ is an inverse of $h = X^m u$.

a) $\implies$ c): let $hh' = 1$ in $A((X))$. Then $v(h) + v(h') = v(1) = 0$ by (3.32.4) c), so $v(h') = -m$, and the coefficient of X^0 in hh' is $h_m h'_{-m} = 1$, so $h_m \in A^\star$. ∎

Proposition 3.32.15 ($k((X))$ is a Field of Fractions for $k[[X]]$)
Let k be a field. Then every element of the field $k((X))$ is a/b for $a, b \in k[[X]]$, $b \neq 0$ – that is, $k((X))$ is a field of fractions for the subring $k[[X]] \subset k((X))$. Moreover, the order function $v : k((X))^\star \to \mathbb{Z}$ is a group homomorphism under multiplication: $v(fg) = v(f) + v(g)$, and $v(f^{-1}) = -v(f)$.

Proof. By (3.32.14), as k is a field every $c \neq 0$ in $k((X))$ is $c = X^m u$ with $m = v(c)$ and $u \in k[[X]]^\star$, which is $uX^m \in k[[X]]$ if $m \geq 0$ and u/X^{-m} otherwise, in either case a ratio of two elements of $k[[X]]$.

By (3.32.4) c), as k is a domain, $v(fg) = v(f) + v(g)$ for $f, g \neq 0$. Applying this to f, f^{-1}: $0 = v(1) = v(ff^{-1}) = v(f) + v(f^{-1})$, giving $v(f^{-1}) = -v(f)$. ∎

Proposition 3.32.16 (Formal Power Series Ring is a DVR)
Let k be a field. Then $k[[X]]$ is an integral domain, not a field, in which every nonzero element a can be written uniquely as $a = X^{v(a)}u$ with $u \in k[[X]]^\star$ (equivalently, $u(0) \neq 0$ (3.32.13)).

Using the criteria (3.30.2)e) $k[[X]]$ is a discrete valuation ring with valuation precisely the order function v.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$k[[X]]$ is a domain, as a subring of the domain $k((X))$ ((3.32.4) c)), and not a field, as $X \neq 0$ has no inverse ($v(Xg) = 1 + v(g) \neq 0$ for every $g \in k[[X]]$ by the same result). Given $0 \neq a \in k[[X]]$, by (3.32.14) $a = X^{v(a)}u$ with $u \in k[[X]]^\star$, where $v(a) \geq 0$: this is exactly the uniformised-element criterion (3.30.2)e), with $\pi := X$. If also $a = X^n u'$ with $u' \in k[[X]]^\star$ then $v(u') = 0$ and $v(a) = n + v(u') = n$ by (3.32.4) c), so the exponent is $v(a)$ and then $u' = u$ as $k((X))$ is a domain. ∎

Proposition 3.32.17 (Coefficient Expansion of a DVR with a Coefficient Field)
Let $(A, \mathfrak{m})$ be a DVR with fraction field K and valuation v, and π a uniformiser of A. Let k be a field and suppose A is a k-algebra with $k(\mathfrak{m}) = k$. Then, for every $n \geq 0$ and $a \in A$, there is a unique $(c_0, \ldots, c_n) \in k^{n+1}$ with

$$a - \sum_{i=0}^{n} c_i \pi^i \in \mathfrak{m}^{n+1},$$

and, writing $c_i(a) \in k$ for the resulting (compatible, as n varies) sequence, there is an injective k-algebra homomorphism

$$L_\pi : A \to k[[X]], \quad L_\pi(a) := \sum_{i\geq 0} c_i(a) X^i,$$

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with $L_\pi(\pi) = X$. Moreover $v_X(L_\pi(a)) = v(a)$ for every $0 \neq a \in A$, and L_π extends uniquely to an injective ring homomorphism $L_\pi : K \to k((X))$ (still denoted L_π).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Existence and uniqueness of $c_i(a)$. By induction on n. For $n = 0$: as $k(\mathfrak{m}) = k$, there is a unique $c_0 \in k$ with $a \equiv c_0 \pmod{\mathfrak{m}}$, that is $a - c_0 \in \mathfrak{m} = \mathfrak{m}^{0+1}$. For the inductive step, given the unique $(c_0, ..., c_{n-1}) \in k^n$ with $b := a - \sum_{i=0}^{n-1} c_i\pi^i \in \mathfrak{m}^n = (\pi^n)$: write $b = \pi^n b'$ for a (unique, π^n a nonzerodivisor) $b' \in A$; by the $n = 0$ case applied to b', there is a unique $c_n \in k$ with $b' - c_n \in \mathfrak{m}$, so $b - c_n\pi^n = \pi^n(b' - c_n) \in \pi^n\mathfrak{m} = \mathfrak{m}^{n+1}$, that is $a - \sum_{i=0}^{n} c_i\pi^i \in \mathfrak{m}^{n+1}$: existence. For uniqueness: if $(c_0, ..., c_n)$ and $(c_{0'}, ..., c_{n'})$ both satisfy the displayed condition, then, dropping the last term $(c_n\pi^n, c_{n'}\pi^n \in \mathfrak{m}^n)$, both $(c_0, ..., c_{n-1})$ and $\left(c_{0'}, ..., c_{(n-1)'}\right)$ satisfy it for $n-1$, so are equal by the inductive hypothesis; then $(c_n - c_{n'})\pi^n = \left(a - \sum_{i<n} c_i\pi^i - c_n\pi^n\right) - \left(a - \sum_{i<n} c_i\pi^i - c_{n'}\pi^n\right) \in \mathfrak{m}^{n+1} = (\pi^{n+1})$, so $c_n - c_{n'} \in (\pi) = \mathfrak{m}$ (π^n a nonzerodivisor); as $k \to A$ is injective (k a field) and $k(\mathfrak{m}) = k$, $c_n - c_{n'} \in k \cap \mathfrak{m} = 0$: $c_n = c_{n'}$. This uniqueness argument shows the tuple for n restricts to the tuple for $n-1$, so the $(c_0, ..., c_n)$ assemble into a well-defined compatible sequence $(c_i(a))_{i\geq 0}$.

L_π is a k-algebra homomorphism. Fix $n \geq 0$. By (3.7.6) (with $A := k$, $B := A/\mathfrak{m}^{n+1}$, $b := \overline{\pi}$), there is a unique k-algebra homomorphism $k[X] \to A/\mathfrak{m}^{n+1}$, $X \mapsto \overline{\pi}$; as $\pi^{n+1} \in \mathfrak{m}^{n+1}$, this kills X^{n+1}, so factors as a k-algebra homomorphism $\overline{\psi}_n : k[X]/(X^{n+1}) \to A/\mathfrak{m}^{n+1}$, $X \mapsto \overline{\pi}$. It is surjective, as $A/\mathfrak{m}^{n+1}$ is spanned, as a k-vector space, by $1, \overline{\pi}, ..., \overline{\pi}^n$ (existence, above); and injective, as $\sum_{i=0}^{n} d_iX^i \in \ker\overline{\psi}_n$ means $\sum_i d_i\pi^i \in \mathfrak{m}^{n+1}$, so $d_i = 0$ for every i (uniqueness, above, applied to $a := 0$). So $\overline{\psi}_n$ is a k-algebra isomorphism, and its inverse $\phi_n := \overline{\psi}_n^{-1}$ satisfies $\phi_n(\overline{a}) = \sum_{i=0}^{n} c_i(a)X^i \mod X^{n+1}$ for every $a \in A$ (as $\overline{\psi}_n$ of the right side is $\overline{a}$, by the defining property of $c_0(a), ..., c_n(a)$).

For $a, b \in A$ and $i \leq n$: comparing coefficients of X^i in $\phi_n\left(\overline{a+b}\right) = \phi_n(\overline{a}) + \phi_n\left(\overline{b}\right)$ (ϕ_n additive) gives $c_i(a+b) = c_i(a) + c_i(b)$, and in $\phi_n\left(\overline{ab}\right) = \phi_n(\overline{a})\phi_n\left(\overline{b}\right)$ (ϕ_n multiplicative) gives $c_i(ab) = \sum_{j=0}^{i} c_j(a)c_{i-j}(b)$; as $n \geq i$ was arbitrary, these hold for every $i \geq 0$. So $L_\pi(a+b) = L_\pi(a) + L_\pi(b)$ and $L_\pi(ab) = L_\pi(a)L_\pi(b)$ (matching the usual formal power series operations on coefficient sequences); and $L_\pi(1) = 1$ (as $1 - 1 \cdot \pi^0 = 0 \in \mathfrak{m}^{n+1}$ for every n, uniqueness gives $c_0(1) = 1$, $c_i(1) = 0$ for $i \geq 1$): L_π is a ring homomorphism. For $c \in k \subset A$: $c - c \cdot \pi^0 = 0 \in \mathfrak{m}^{n+1}$ for every n, so $c_0(c) = c$, $c_i(c) = 0$ for $i \geq 1$, giving $L_\pi(c) = c$ (the constant series): L_π is k-linear, hence a k-algebra homomorphism.

$L_\pi(\pi) = X$. For every $n \geq 1$, $(0, 1, 0, ..., 0) \in k^{n+1}$ satisfies $\pi - (0 \cdot \pi^0 + 1 \cdot \pi^1 + 0 + \cdots + 0) = 0 \in \mathfrak{m}^{n+1}$ (and $(0) \in k^1$ satisfies it for $n = 0$, as $\pi \in \mathfrak{m}$); by uniqueness, $c_0(\pi) = 0$, $c_1(\pi) = 1$, $c_i(\pi) = 0$ for $i \geq 2$, so $L_\pi(\pi) = X$.

Valuation formula and injectivity. Let $0 \neq a \in A$, $m := v(a)$, so $a = u\pi^m$ for a unit $u \in A^\star$ (3.30.2). For $i < m$: $a \in \mathfrak{m}^m \subseteq \mathfrak{m}^{i+1}$, so $(0, ..., 0) \in k^{i+1}$ satisfies the defining property for a at level i; by uniqueness, $c_j(a) = 0$ for every $j \leq i < m$. For $i = m$: writing $\overline{u} \in k$ for the unique element with $u - \overline{u} \in \mathfrak{m}$ (as $k(\mathfrak{m}) = k$; and $\overline{u} \neq 0$, as $u \notin \mathfrak{m}$), $a - \overline{u}\pi^m = \pi^m(u - \overline{u}) \in \pi^m\mathfrak{m} = \mathfrak{m}^{m+1}$, so, with $c_0(a) = \cdots = c_{m-1}(a) = 0$ already fixed, uniqueness gives $c_m(a) = \overline{u} \neq 0$. So $L_\pi(a) = \overline{u}X^m + \sum_{i>m} c_i(a)X^i$, giving $v_X(L_\pi(a)) = m = v(a)$; in particular $L_\pi(a) \neq 0$ for $a \neq 0$, so L_π is injective.

L_π extends to K. As $L_\pi : A \to k[[X]]$ is injective (above) and $k[[X]] \subset k((X))$ (a field) (3.32.15), $A \setminus \{0\}$ maps into $k((X))^\star$. So (3.5.5) (with $S := A \setminus \{0\}$, so $S^{-1}A = K$) gives a unique ring homomorphism $L_\pi : K \to k((X))$ (still denoted L_π) with $L_\pi|_A$ the original map, explicitly $L_\pi(a/s) = L_\pi(a)L_\pi(s)^{-1}$; it is injective (if $L_\pi(a/s) = 0$ then $L_\pi(a) = 0$, so $a = 0$, giving $a/s = 0$). ∎

Proposition 3.32.18 (Product Rule)
Let A be a commutative ring and $f, g \in A((X))$. Then $(fg)' = f'g + fg'$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $f = \sum_i a_iX^i$, $g = \sum_j b_jX^j$ and fix $n \in \mathbb{Z}$. All sums below are over $i \in \mathbb{Z}$ and are sums of families with all but finitely many terms zero, with the coefficients of a product as in (3.32.3). The coefficient of X^{n-1} in $(fg)'$ is $n(fg)_n = n\sum_i a_ib_{n-i}$. Since $(f')_k = (k+1)a_{k+1}$ and $(g')_k = (k+1)b_{k+1}$, reindexing $i = k+1$ and $i = n-k-1$ respectively ((3.2.53)) gives

$$(f'g)_{n-1} = \sum_i ia_ib_{n-i} \qquad (fg')_{n-1} = \sum_i (n-i)a_ib_{n-i}$$

Hence

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$$\begin{aligned}(f'g+fg')_{n-1} &= \sum_i i a_i b_{n-i} + \sum_i (n-i) a_i b_{n-i} && \text{by the above}\\ &= \sum_i n a_i b_{n-i} && \text{by (3.2.51)}\\ &= n\sum_i a_i b_{n-i} = ((fg)')_{n-1} && \text{by (3.3.17)}\end{aligned}$$

As n was arbitrary, $(fg)' = f'g + fg'$. ∎

Lemma 3.32.19 (Power Rule)
Let A be a commutative ring and $\psi \in A((X))$. Then $(\psi^n)' = n\psi^{n-1}\psi'$ for every $n \geq 0$. If moreover ψ is a unit of $A((X))$, this also holds for every $n < 0$.

Proof. **$n \geq 0$.** Induction on n: the case $n = 0$ is $0 = 0$ ($\psi^0 = 1$ is constant). Given $(\psi^n)' = n\psi^{n-1}\psi'$, the product rule (3.32.18) gives

$$(\psi^{n+1})' = (\psi \cdot \psi^n)' = \psi'\psi^n + \psi(\psi^n)' = \psi'\psi^n + n\psi^n\psi' = (n+1)\psi^n\psi'$$

$n < 0$, ψ a unit. Differentiating $\psi\psi^{-1} = 1$ with the product rule, $0 = \psi'\psi^{-1} + \psi(\psi^{-1})'$, so $(\psi^{-1})' = -\psi^{-2}\psi'$ – the case $n = -1$. For $n \leq -2$, write $n = -r$, $r \geq 2$: applying the $n \geq 0$ case just shown to the element $\psi^{-1} \in A((X))$ (no hypothesis on its order was used there),

$$(\psi^{-r})' = \left((\psi^{-1})^r\right)' = r(\psi^{-1})^{r-1}(\psi^{-1})' = r(\psi^{-1})^{r-1}(-\psi^{-2}\psi') = -r\psi^{-r-1}\psi' = n\psi^{n-1}\psi'$$

∎

Proposition 3.32.20 (Substitution into a Formal Laurent Series)
Let A be an integral domain, $f = \sum_{n \geq n_0} c_n X^n \in A((X))$ and $\psi \in XA[[X]]$ (that is, $\psi(0) = 0$). Suppose that either

- *$f \in A[[X]]$, that is $n_0 \geq 0$, or*
- *$\psi \in A((X))^\star$, that is $\psi \in X^m A[[X]]^\star$ for some $m > 0$ (3.32.14)*

Then

$$(f \circ \psi)(X) := \sum_{n \geq n_0} c_n \psi(X)^n$$

is a well-defined element of $A((X))$ (with ψ^n, $n < 0$, the power in the ring $A((X))$), and $f \circ \psi \in A[[X]]$ if $f \in A[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\psi \in XA[[X]]$ has no terms of degree ≤ 0 we have $v(\psi) \geq 1$, so $v(\psi^n) \geq n$ for every $n \geq 0$ by (3.32.4) b). Negative powers of ψ only occur for the finitely many n with $n_0 \leq n < 0$. Hence for every N all $n \geq \max(N, 0)$ satisfy $v(c_n\psi^n) \geq n \geq N$, and only the finitely many n with $n_0 \leq n < \max(N, 0)$ can fail this, so $(c_n\psi^n)_{n \geq n_0}$ is summable (3.32.9) and its sum lies in $A((X))$. If $f \in A[[X]]$ then every $c_n\psi^n \in A[[X]]$ and the sum lies in $A[[X]]$ by (3.32.10) a). ∎

Proposition 3.32.21 (Chain Rule)
In the setting of (3.32.20), $(f \circ \psi)' = (f' \circ \psi) \cdot \psi'$.

Proof. By the power rule ((3.32.19), which for $n < 0$ requires that ψ is a unit of $A((X))$, as it is then), $(c_n\psi^n)' = c_n(\psi^n)' = nc_n\psi^{n-1}\psi'$ for every $n \geq n_0$, using linearity of $(-)'$ (immediate from (3.32.7), as $c_n \in A$ is a constant). The term $n = 0$ is 0, and is simply omitted if ψ is not a unit. Then, by termwise differentiation (3.32.10) c) of the summable family $(c_n\psi^n)_{n \geq n_0}$ ((3.32.20)'s proof),

$$(f \circ \psi)' = \sum_{n \geq n_0} (c_n\psi^n)' = \sum_{n \geq n_0} nc_n\psi^{n-1}\psi'$$

The family $(nc_n\psi^{n-1})_{n \geq n_0}$ is summable by the same order bound as $(c_n\psi^n)_{n \geq n_0}$ ((3.32.20)'s proof, shifted down one degree), so distributivity ((3.32.10) d)) gives

$$\sum_{n \geq n_0} nc_n\psi^{n-1}\psi' = \left(\sum_{n \geq n_0} nc_n\psi^{n-1}\right)\psi'$$

Reindexing $m = n - 1$, $\sum_{n \geq n_0} nc_n\psi^{n-1} = \sum_{m \geq n_0 - 1}(m+1)c_{m+1}\psi^m = f' \circ \psi$, giving $(f \circ \psi)' = (f' \circ \psi) \cdot \psi'$. ∎

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Proposition 3.32.22 (Substitution is a Ring Homomorphism)
Let A be an integral domain and $\psi \in XA[[X]]$. Then $f \mapsto f \circ \psi$ is a ring homomorphism $A[[X]] \to A[[X]]$. If moreover the leading coefficient of ψ is a unit (so $\psi \in A((X))^\star$ by (3.32.14)) then $f \mapsto f \circ \psi$ is a ring homomorphism $A((X)) \to A((X))$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The arguments below apply to f, g, h in $A[[X]]$ in the first case and in $A((X))$ in the second.

Identity. $1 \circ \psi = \psi^0 = 1$.

Additivity. Write $f = \sum_{n \geq n_0} c_n X^n$, $h = \sum_{n \geq m_0} d_n X^n$, and set $p_0 := \min(n_0, m_0)$, extending both coefficient sequences by $c_n := 0$ (resp. $d_n := 0$) for $p_0 \leq n < n_0$ (resp. $< m_0$); this changes neither series, nor the sums below, as it only adds zero terms. Then $f + h = \sum_{n \geq p_0} (c_n + d_n) X^n$, and the families $(c_n \psi^n)_{n \geq p_0}$ and $(d_n \psi^n)_{n \geq p_0}$ are summable by (3.32.20). Hence

$$\begin{aligned}(f+h) \circ \psi &= \sum_{n \geq p_0} (c_n + d_n)\psi^n = \sum_{n \geq p_0} (c_n \psi^n + d_n \psi^n) && \text{by distributivity in } A((X)) \\ &= \sum_{n \geq p_0} c_n \psi^n + \sum_{n \geq p_0} d_n \psi^n = f \circ \psi + h \circ \psi && \text{by (3.32.10) } e)\end{aligned}$$

Multiplicativity. Write $f = \sum_{i \geq i_0} c_i X^i$, $g = \sum_{j \geq j_0} d_j X^j$, so $f \circ \psi = \sum_i c_i \psi^i$ and $g \circ \psi = \sum_j d_j \psi^j$, and let $P := \mathbb{Z}_{\geq i_0} \times \mathbb{Z}_{\geq j_0}$ and $a(i,j) := c_i d_j \psi^{i+j}$ for $(i,j) \in P$. We have $v(a(i,j)) \geq i + j$ whenever $i + j \geq 0$ as $v(\psi^n) \geq n$ for $n \geq 0$ ((3.32.20)'s proof), and for every N only the finitely many pairs with $i + j < \max(N, 0)$ can fail $v(a(i,j)) \geq N$, so $(a(i,j))_{(i,j) \in P}$ is summable. Then

$$\begin{aligned}(f \circ \psi)(g \circ \psi) &= \sum_{j \geq j_0} (f \circ \psi) d_j \psi^j && \text{by (3.32.10) } d) \\ &= \sum_{j \geq j_0} \sum_{i \geq i_0} c_i d_j \psi^{i+j} && \text{by (3.32.10) } d) \\ &= \sum_{(i,j) \in P} a(i,j) && \text{by (3.32.11)}\end{aligned}$$

and, grouping instead by $N := i + j$, whose fibres are finite,

$$\begin{aligned}(fg) \circ \psi &= \sum_N \left(\sum_{i+j=N} c_i d_j \right) \psi^N = \sum_N \sum_{i+j=N} c_i d_j \psi^N && \text{by (3.3.17)} \\ &= \sum_{(i,j) \in P} a(i,j) && \text{by (3.32.10) } f) \text{ for the partition of } P \text{ by } i+j\end{aligned}$$

so $(f \circ \psi)(g \circ \psi) = (fg) \circ \psi$. ∎

3.32.1 Exponential and Logarithm

Definition 3.32.23 (Formal Exponential and Logarithm)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Define

$$\exp(X) := \sum_{n \geq 0} \frac{X^n}{n!} \in A[[X]] \qquad \log(1+X) := \sum_{n \geq 1} (-1)^{n+1} \frac{X^n}{n} \in XA[[X]]$$

For $f \in XA[[X]]$ and $g \in 1 + XA[[X]]$ define $\exp(f) := \exp(X) \circ f$ and $\log(g) := \log(1+X) \circ (g-1)$, using substitution (3.32.20) (note $g - 1 \in XA[[X]]$).

Lemma 3.32.24 (Zero Derivative)
Let A be a commutative ring with $\mathbb{Q} \subseteq A$ and $F \in A((X))$ with $F' = 0$. Then F is a constant, that is $F \in A$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$(F')_n = (n+1)F_{n+1} = 0$ for every $n \in \mathbb{Z}$, and $n + 1$ is invertible in A for $n \neq -1$, so $F_{n+1} = 0$ for all $n \neq -1$, that is $F = F_0$. ∎

Proposition 3.32.25 (Derivatives of exp and log)
Let A be an integral domain with $\mathbb{Q} \subseteq A$, $f \in XA[[X]]$ and $g \in 1 + XA[[X]]$. Then $\exp(f) \in 1 + XA[[X]]$ and $\log(g) \in XA[[X]]$, and

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$$\exp(f)' = \exp(f)f' \qquad \log(g)' = g'g^{-1}$$

where g is a unit of $A[[X]]$ by (3.32.13).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

exp. Directly $\exp(X)' = \sum_{n\geq 1} \frac{X^{n-1}}{(n-1)!} = \exp(X)$. Since $v(f^n) \geq n$, only the term $n = 0$ contributes to the constant coefficient of $\exp(f) = \sum_n \frac{f^n}{n!}$, so $\exp(f) \in 1 + XA[[X]]$. By the chain rule (3.32.21) $\exp(f)' = (\exp' \circ f)f' = \exp(f)f'$.

log. Likewise $\log(g) \in XA[[X]]$, as each term of $\sum_{n\geq 1} (-1)^{n+1}\frac{(g-1)^n}{n}$ has order at least n. Directly $\log(1+X)' = \sum_{n\geq 1}(-1)^{n+1}X^{n-1} = \sum_{m\geq 0}(-X)^m = (1+X)^{-1}$, as $(1+X)\sum_{m\geq 0}(-X)^m = 1$ coefficientwise. By the chain rule

$$\log(g)' = (\log(1+X)' \circ (g-1))g' = ((1+X)^{-1} \circ (g-1))g' = g^{-1}g'$$

since $f \mapsto f \circ (g-1)$ is a ring homomorphism (3.32.22) and $(1+X)\circ(g-1) = g$. ∎

Proposition 3.32.26 (Functional Equations of exp and log)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Then $\exp(f+g) = \exp(f)\exp(g)$ for $f, g \in XA[[X]]$, and $\log(gh) = \log(g) + \log(h)$ for $g, h \in 1 + XA[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

exp. Write $E := \exp(f+g)$ and $U := \exp(f)\exp(g)$, a unit of $A[[X]]$ with constant coefficient 1. By the product rule and (3.32.25), $E' = E(f'+g')$ and $U' = U(f'+g')$. Hence for $R := EU^{-1}$, using the power rule (3.32.19) for U^{-1},

$$\begin{aligned} R' &= E'U^{-1} - EU'U^{-2} && \text{by the product and power rules} \\ &= E(f'+g')U^{-1} - EU(f'+g')U^{-2} = 0 \end{aligned}$$

so R is constant by (3.32.24), equal to $R(0) = 1$, that is $E = U$.

log. By the product rule and (3.32.25), $\log(gh)' = (g'h + gh')g^{-1}h^{-1} = g'g^{-1} + h'h^{-1} = (\log(g) + \log(h))'$. So $\log(gh) - \log(g) - \log(h)$ has derivative 0 and constant coefficient 0, hence is 0 by (3.32.24). ∎

Proposition 3.32.27 (exp and log are Inverse Isomorphisms)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Then $\log(\exp(f)) = f$ for $f \in XA[[X]]$ and $\exp(\log(g)) = g$ for $g \in 1 + XA[[X]]$. Hence exp and log are mutually inverse isomorphisms of groups between $(XA[[X]], +)$ and $(1 + XA[[X]], \times)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.32.25) $\log(\exp(f))' = \exp(f)'\exp(f)^{-1} = f'$, so $\log(\exp(f)) - f$ has derivative 0 and constant coefficient 0, and is 0 by (3.32.24). For the other composite let $E := \exp(\log(g))$, so $E' = E\log(g)' = Eg'g^{-1}$ by the same result. Then

$$\begin{aligned} (Eg^{-1})' &= E'g^{-1} - Eg'g^{-2} && \text{by the product and power rules} \\ &= Eg'g^{-2} - Eg'g^{-2} = 0 \end{aligned}$$

so Eg^{-1} is constant by (3.32.24), equal to $E(0)g(0)^{-1} = 1$, that is $E = g$. Finally exp is a homomorphism by (3.32.26), with inverse log. ∎

3.32.2 Extensions of Power Series Rings

The main goal of this subsection is to show that, for k algebraically closed, the valuation of $k((t))$ has a **unique** extension to any finite separable extension E of $k((t))$, and that E is itself a power series field $k((u))$: see (3.32.33) and (3.32.34). The main tool is the Weierstrass division lemma for formal power series, which we record first.

Remark 3.32.28
*[Che51] (Ch. II, §6) treats the case of (3.32.33) where the ramification index e is prime to $\mathrm{char}(k)$, obtaining the sharper conclusion that t itself is an e-th power in $k((u))$ (a literal Puiseux expansion $k\big(\big(t^{1/e}\big)\big)$), via Hensel's lemma applied to $W^e - v$ for the unit $v := t/u^e$. That argument is unavailable when $\mathrm{char}(k) = p \mid e$: the derivative eW^{e-1} vanishes identically, so $W^e - v$ has no simple root to lift, and t need not be an e-th power at all (an Artin-Schreier extension gives a counterexample). (3.32.33) instead only produces $t = u^e c$ for **some** unit c, obtained by solving $\tau G(\tau) = u^e$ by direct coefficient-stabilisation rather than root extraction; this sidesteps the characteristic obstruction entirely, at the cost of the sharper radical form, which is in any case not needed below.*

Lemma 3.32.29 (Division with Remainder by a Distinguished Power Series)
Let R be a commutative ring, $m \geq 1$ an integer, $w(X) \in R[[X]]^\star$, and $\tau(X) := X^m w(X)$. Then every $p \in R[[X]]$ is uniquely $p = q\tau + r$ with $q \in R[[X]]$ and $r \in R[X]$ of degree $< m$. In particular τ is a non-zero-divisor in $R[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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For existence, split $p = X^m p' + r$ with r the truncation below degree m and p' the shift, and put $q := p' w^{-1}$ (as $w \in R[[X]]^\star$ (3.32.13)). For uniqueness, if $q\tau + r = 0$ then $r = -q\tau \in X^m R[[X]]$ has degree $< m$, so $r = 0$; then $qX^m w = 0$ gives $q = 0$, as w is a unit and X^m is a non-zero-divisor. In particular if $q\tau = 0$ then $q = 0$, that is τ is a non-zero-divisor. ∎

Lemma 3.32.30 (Free Basis for a Distinguished Power Series)
Let R be a commutative ring, $m \geq 1$ an integer, $w(X) \in R[[X]]^\star$, and $\tau(X) := X^m w(X) \in R[[X]]$ (so the coefficients of $\tau(X)$ below degree m vanish, and its coefficient of X^m is $w(0) \in R^\star$). Then substitution $Y \mapsto \tau(X)$ (well-defined as $\tau(X)$ has zero constant term (3.32.5)) is an injective ring homomorphism $R[[Y]] \to R[[X]]$, $f(Y) \mapsto f(\tau(X))$, and $R[[X]]$ is a free $R[[Y]]$-module of rank m with basis $1, X, \ldots, X^{m-1}$. Explicitly, every $p \in R[[X]]$ is uniquely

$$p(X) = \sum_{i=0}^{m-1} f_i(\tau(X)) X^i$$

for power series $f_0, \ldots, f_{m-1} \in R[[Y]]$. (In the proof we abbreviate $w(X)$, $\tau(X)$ and $p(X)$ to w, τ and p.)

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Existence. Put $p_0 := p$ and, dividing repeatedly by τ (3.32.29), write $p_j = p_{j+1}\tau + r_j$ with $\deg r_j < m$. Then for every $J \geq 0$

$$p = p_{J+1}\tau^{J+1} + \sum_{j=0}^{J} r_j \tau^j, \quad \text{with } p_{J+1}\tau^{J+1} \in X^{m(J+1)} R[[X]].$$

This follows by induction on J: for $J = 0$ it is $p = p_1\tau + r_0$, and substituting $p_{J+1} = p_{J+2}\tau + r_{J+1}$ gives the step, as $\tau^{J+1} \in X^{m(J+1)}R[[X]]$ because $\tau = X^m w$. So $\left(r_j\tau^j\right)_{j\geq 0}$ is summable (3.32.9) (as $v\left(r_j\tau^j\right) \geq mj$), and $p = \sum_{j\geq 0} r_j\tau^j$ by (3.32.12), as $p - \sum_{j\leq J} r_j\tau^j = p_{J+1}\tau^{J+1} \in X^{m(J+1)}R[[X]]$ for every J. Write $r_j = \sum_{i=0}^{m-1} \gamma_{j,i} X^i$ and set $f_i := \sum_{j\geq 0} \gamma_{j,i} Y^j \in R[[Y]]$. The family $\left(\gamma_{j,i}\tau^j X^i\right)_{(j,i)}$ is summable, as $v\left(\gamma_{j,i}\tau^j X^i\right) \geq mj$. Then

$$\begin{aligned}
\sum_{i=0}^{m-1} f_i(\tau) X^i &= \sum_{i=0}^{m-1} \sum_{j\geq 0} \gamma_{j,i}\tau^j X^i \quad \text{by the definition of } f_i(\tau) \text{ and (3.32.10) } d) \\
&= \sum_{j\geq 0} \sum_{i=0}^{m-1} \gamma_{j,i}\tau^j X^i \quad \text{by (3.32.11)} \\
&= \sum_{j\geq 0} r_j \tau^j = p.
\end{aligned}$$

Uniqueness. Suppose $\sum_{i=0}^{m-1} f_i(\tau) X^i = 0$; we show $f_i \in Y^N R[[Y]]$ for all i and all $N \geq 0$, whence every $f_i = 0$ (3.32.6) (in $R[[Y]]$). The case $N = 0$ is trivial. Suppose $f_i = Y^N g_i$ for all i. Substitution is a ring homomorphism (3.32.22), so $f_i(\tau) = \tau^N g_i(\tau)$ and $\tau^N \sum_i g_i(\tau) X^i = 0$; as τ is a non-zero-divisor (3.32.29), $\sum_i g_i(\tau) X^i = 0$. Write $g_i = c_i + Y h_i$ with $c_i \in R$. Then

$$0 = \sum_{i=0}^{m-1} g_i(\tau) X^i = \left(\sum_{i=0}^{m-1} h_i(\tau) X^i \right) \tau + \sum_{i=0}^{m-1} c_i X^i,$$

which is a division of 0 by τ, so the uniqueness in (3.32.29) gives $c_i = 0$ for every i. Hence $f_i = Y^{N+1} h_i$ for every i, completing the induction.

Module structure. Substitution $f \mapsto f(\tau)$ is a ring homomorphism (3.32.22), and it is injective: if $f(\tau) = 0$ then $f(\tau)X^0 + \sum_{i=1}^{m-1} 0 \cdot X^i = 0$, so uniqueness (with $p = 0$) forces $f = 0$. Regarding $R[[X]]$ as an $R[[Y]]$-module through it, $f \cdot p = f(\tau)p$, existence says $1, X, \ldots, X^{m-1}$ generate, and uniqueness says they are linearly independent. ∎

Example 3.32.31
Take $w = 1$, so $\tau = X^m$. For $p = \sum_{n\geq 0} a_n X^n \in R[[X]]$, writing each $n = mj + i$ with $0 \leq i < m$ gives

$$p = \sum_{i=0}^{m-1} \left(\sum_{j\geq 0} a_{mj+i} X^{mj} \right) X^i, \quad \text{so} \quad f_i(Y) = \sum_{j\geq 0} a_{mj+i} Y^j.$$

That is, f_i collects the coefficients of p whose exponent is congruent to i modulo m. For $m = 2$, f_0 collects the even-index coefficients and f_1 the odd-index ones.

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Corollary 3.32.32 (Specialisation Compatibility)
In the setting of (3.32.30), let $\phi : R \to S$ be a homomorphism of commutative rings, extended coefficientwise to $\phi : R[[X]] \to S[[X]]$. Then $\phi(w) \in S[[X]]^\star$, and the decomposition of $\phi(p)$ relative to $\phi(\tau) = X^m\phi(w)$ is $\phi(p) = \sum_{i=0}^{m-1} \phi(f_i)(\phi(\tau))X^i$, where $\phi(f_i) \in S[[Y]]$ denotes ϕ applied coefficientwise to f_i.

Proof. If $wv = 1$ then $\phi(w)\phi(v) = 1$, so $\phi(w) \in S[[X]]^\star$. Next, $\phi(f(\tau)) = \phi(f)(\phi(\tau))$ for every $f \in R[[Y]]$. Fix $N \geq 0$ and write $f = s + Y^N g$ with $s \in R[Y]$ of degree $< N$ and $g \in R[[Y]]$. Then $f(\tau) = s(\tau) + \tau^N g(\tau)$ (the tail $\sum_{n \geq N} c_n\tau^n = \tau^N \sum_n c_{n+N}\tau^n$, (3.32.9)), and likewise $\phi(f) = \phi(s) + Y^N\phi(g)$, so $\phi(f)(\phi(\tau)) = \phi(s)(\phi(\tau)) + \phi(\tau)^N\phi(g)(\phi(\tau))$. Applying the ring homomorphism ϕ to $f(\tau) = s(\tau) + \tau^N g(\tau)$ gives $\phi(f(\tau)) = \phi(s)(\phi(\tau)) + \phi(\tau)^N\phi(g(\tau))$, so $\phi(f(\tau)) - \phi(f)(\phi(\tau)) \in \phi(\tau)^N S[[X]] \subseteq X^{mN}S[[X]]$, as $\phi(\tau) = X^m\phi(w)$. This holds for every N and $\bigcap_N X^N S[[X]] = 0$ (3.32.6), so $\phi(f(\tau)) = \phi(f)(\phi(\tau))$. Applying ϕ to $p = \sum_i f_i(\tau)X^i$ therefore gives $\phi(p) = \sum_i \phi(f_i)(\phi(\tau))X^i$. This exhibits $\phi(p)$ in the form guaranteed unique by (3.32.30) over S, so it is **the** decomposition of $\phi(p)$. ∎

Lemma 3.32.33 (Finite Separable Extensions of $k((t))$ are Power Series Fields)
Let k be algebraically closed, $F := k((t))$, and E/F a finite separable extension of degree n. Then there are $u \in E$, an element $\tau \in uk[[u]]$ with $\tau = u^n c$ for some $c \in k[[u]]^\star$, and a field isomorphism $\Psi : E \xrightarrow{\sim} k((u))$ with $\Psi|_F$ the substitution $t \mapsto \tau$ and $\Psi(u) = u$. In particular $1, u, \ldots, u^{n-1}$ is an F-basis of E.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Step 1: a DVR. Let $\mathcal{O}_F := k[[t]]$, a DVR with fraction field F (3.32.16) (so a Dedekind domain (3.31.3)), and let C be the integral closure of $\mathcal{O}_F$ in E. By (3.23.26) C is a finite $\mathcal{O}_F$-module, and by (3.31.11) it is Dedekind. By lying-over (3.23.17) there is a prime $\mathfrak{p} \subset C$ with $\mathfrak{p} \cap \mathcal{O}_F = (t)$. Put $B := C_{\mathfrak{p}}$, a DVR with fraction field E, and let $e := v_B(t) \geq 1$.

Step 2: residue field and valuation. $C/\mathfrak{p}$ is a domain, finite over $\mathcal{O}_F/(t) = k$, so $k \hookrightarrow C/\mathfrak{p}$ is injective and integral, hence $C/\mathfrak{p}$ is a field (3.23.14), finite over k. As $B = C_{\mathfrak{p}}$, (3.5.43) gives $B/\mathfrak{m}_B \cong \mathrm{Frac}(C/\mathfrak{p}) = C/\mathfrak{p}$ ($C/\mathfrak{p}$ already a field); as k is algebraically closed, $k \to B/\mathfrak{m}_B$ is an isomorphism. Every $a \in \mathcal{O}_F^\times$ is a unit of B, so $v_B(x) = ev_F(x)$ for $x \in F^\times$. Let u be a uniformiser of B, so $t = u^e w$ with $w \in B^\star$.

Step 3: expansions. Every $b \in B$ can be written $b = \sum_{i<e} f_i u^i$ with $f_i \in k[[t]]$. We show by induction on $N \geq 0$ that $b \equiv \sum_{i<e}\left(\sum_{j<N} c_{ij}t^j\right)u^i \pmod{t^N B}$ for some $c_{ij} \in k$. The case $N = 0$ is trivial. For the inductive step, write the current remainder as $t^N y$ with $y \in B$ (inductive hypothesis); since $B/uB = k$, repeated division by u gives $y = \sum_{i<e} d_i u^i + u^e y'$ with $d_i \in k$, $y' \in B$ (there is no need to iterate past degree $e-1$: stop as soon as e terms are extracted). Using $u^e = tw^{-1}$,

$$t^N y = \sum_{i<e} (t^N d_i)u^i + t^{N+1}(w^{-1}y'),$$

with $w^{-1}y' \in B$; so taking $c_{i,N} := d_i$ advances the induction, with new remainder $t^{N+1}(w^{-1}y') \in t^{N+1}B$. Put $f_i := \sum_j c_{ij}t^j \in k[[t]]$. Then $z := b - \sum_i f_i u^i$ satisfies $z \in t^N B$ for every N, so $v_B(z) \geq eN$ for all N and $z = 0$.

Step 4: basis and degree. Every element of E has the form $t^{-N}b$ with $b \in B$, so Step 3 gives $E = \sum_{i<e} Fu^i$. These elements are independent: if $\sum_{i<e} f_i u^i = 0$ with $f_i \in F$ not all zero, the valuations $v_B(f_i u^i) = ev_F(f_i) + i$ are distinct modulo e for the nonzero f_i, so the minimum is attained exactly once, a contradiction. Hence $n = e$, and $1, \ldots, u^{e-1}$ is a basis. Moreover $B = \bigoplus_{i<e} k[[t]]u^i$, by the valuation estimate $\min_i(ev_F(f_i) + i) = v_B(b) \geq 0$ applied to $b = \sum_i f_i u^i$ with $f_i \in F$.

Step 5: the minimal polynomial. Write $w^{-1} = \sum_{i<e} g_i u^i$ with $g_i \in k[[t]]$ (Step 3). The constant $g_0(0)$ is the residue of w^{-1}, so $g_0(0) \in k^\star$. From $u^e = tw^{-1}$, u is a root of

$$P(Y) := Y^e - t\sum_{i<e} g_i(t)Y^i \in F[Y].$$

Since $F[u] = E$ has degree e over F, P is the minimal polynomial of u, and $E \cong F[Y]/(P)$ with $Y \mapsto u$.

Step 6: solving for τ. For $s \in uk[[u]]$ put $G(s) := \sum_{i<e} g_i(s)u^i \in k[[u]]$, where $g_i(s)$ is the substitution (well defined as s has zero constant term). Since $G(s) \equiv g_0(0) \pmod{u}$, $G(s)$ is a unit. Define $T(s) := u^e G(s)^{-1} \in u^e k[[u]]$. For $s \equiv s' \pmod{u^N}$ we have $g_i(s) - g_i(s') \in (s - s')k[[u]] \subseteq u^N k[[u]]$, hence $G(s) \equiv G(s') \pmod{u^N}$ and $T(s) \equiv T(s') \pmod{u^{N+e}}$. Starting from $s_0 := 0$ and setting $s_{m+1} := T(s_m)$, the sequence satisfies $s_{m+1} \equiv s_m \pmod{u^{m+e}}$, so each coefficient stabilises. Let $\tau \in uk[[u]]$ be the coefficientwise limit. Then $\tau \equiv T(\tau) \pmod{u^N}$ for every N, so $\tau = T(\tau)$, i.e. $\tau G(\tau) = u^e$. Also $\tau = u^e c$ with $c \in k[[u]]^\star$, since $\tau G(\tau) = u^e$ and $G(\tau)$ is a unit.

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Step 7: the homomorphism. The substitution $t \mapsto \tau$ defines a ring homomorphism $\sigma : k[[t]] \to k[[u]]$ ((3.32.22), as τ has zero constant term), and since τ is a unit times u^e it extends to a field homomorphism $\sigma : F \to k((u))$. The homomorphism $F[Y] \to k((u))$, $Y \mapsto u$, extending σ, kills P:

$$\sigma(P)(u) = u^e - \tau \sum_{i<e} g_i(\tau) u^i = u^e - \tau G(\tau) = 0.$$

So it factors through $E \cong F[Y]/(P)$, giving a field homomorphism $\Psi : E \to k((u))$ with $\Psi|_F = \sigma$ and $\Psi(u) = u$. It is injective, as E is a field.

Step 8: surjectivity. Since $\tau = u^e c$ with $c \in k[[u]]^\star$, the Weierstrass division lemma (3.32.30) (with $R := k$, $X := u$, $m := e$, $w := c$, so the series of the lemma is τ) gives $k[[u]] = \bigoplus_{i<e} \sigma(k[[t]])u^i$. Multiplying by powers of τ gives $k((u)) = \sum_{i<e} \sigma(F)u^i = \sigma(F)[u]$. But $\operatorname{im}\Psi = \Psi(F[u]) = \sigma(F)[u]$, so Ψ is surjective.

Hence $\Psi : E \xrightarrow{\sim} k((u))$ is an isomorphism of fields over F, with $\Psi(u) = u$ and $\Psi(t) = \tau = u^n c$. ∎

Corollary 3.32.34 (Uniqueness of the Extended Valuation)
In the setting of (3.32.33), let C be the integral closure of $k[[t]]$ in E. Then C has exactly one prime $\mathfrak{p}$ lying over (t), the DVR $C_\mathfrak{p}$ is the unique valuation ring of E containing $k[[t]]$ whose maximal ideal contracts to (t), and its ramification index is $[E : F]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\mathfrak{q}$ be any prime of C over (t), which exists by lying-over. Steps 2–4 of the proof of (3.32.33) apply verbatim to the DVR $B := C_\mathfrak{q}$ (which has residue field k), and give $e_\mathfrak{q} = v_B(t) = [E : F] = n$.

Now apply the Ramification Formula (3.31.20) to the AKLB situation $k[[t]] \subset C$ over E/F (both Dedekind by (3.31.11), and E/F separable), at the prime (t):

$$\sum_{\mathfrak{q} \mid (t)} e_\mathfrak{q} f_\mathfrak{q} = [C/tC : k] = n.$$

Each $f_\mathfrak{q} = [C/\mathfrak{q} : k]$ is finite, and $C/\mathfrak{q}$ is a field algebraic over k, hence $f_\mathfrak{q} = 1$. So $\sum_\mathfrak{q} e_\mathfrak{q} = n$. By the previous paragraph each term equals n, so there is exactly one prime $\mathfrak{p}$ over (t) and $e_\mathfrak{p} = n$.

For uniqueness of the valuation ring, let $V \subseteq E$ be a valuation ring with $k[[t]] \subseteq V$ and $\mathfrak{m}_V \cap k[[t]] = (t)$. By (3.24.14), $C \subseteq V$. Then $\mathfrak{m}_V \cap C$ is a prime of C over (t), hence equals $\mathfrak{p}$. Every element of $C \setminus \mathfrak{p}$ lies outside $\mathfrak{m}_V$, so is a unit of V; hence $C_\mathfrak{p} \subseteq V$. Since $C_\mathfrak{p}$ is a valuation ring of E and valuation rings are maximal for domination among proper subrings of E (3.24.2), $V = C_\mathfrak{p}$. ∎

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